ENTRIES 071-072 One Unified Uncertainty Model for All Three Chips — September 4, 2026 — Status: COMPLETE

Retiring the Two-Lineage Split

Kyiv/Sherbrooke Had Calibrated Uncertainty, Brisbane Didn't -- Now All Three Do

Why This Was Needed

Since Brisbane was added (Entry 061), the deployed demo ran two separate model lineages: Kyiv/Sherbrooke used Entry 057's MC-Dropout model, trained before Brisbane existed on a 2-chip, 2,329-circuit dataset, with a validated calibrated uncertainty band. Brisbane used Entry 065/069's combined generalist model, trained without dropout (so any inference-time-only dropout applied to it would have produced an uncertainty-shaped number that was never validated -- the project has been careful throughout not to do this). The result was an honest but visible asymmetry: two different models, two different guarantees, patched around with explicit labeling in the demo rather than actually resolved.

Entry 071 — One Model, Trained With Dropout, On All Current Data

A single MC-Dropout-enabled model (same architecture, same per-chip normalization, same floor-collapse up-weighting as every GNN since Entry 048/051) was trained on the full current entry068_graph_dataset.json -- 5,690 circuits across Kyiv, Sherbrooke, and Brisbane together, the balanced post-Entry-067 dataset. Dropout (rate=0.15) is active both during training and at inference (T=20 stochastic passes), matching Entry 057's original methodology exactly -- only the underlying data and chip count changed.

Calibration Check — Genuine Held-Out Validation

	MAE	R^2	mean predicted std	corr(std, |error|)
Held-out test (80/20 CV)	1.38	0.964	0.81 pts	0.553

This calibration run used a separate model trained on only 80% of the data (never seeing the held-out 20%), exactly mirroring Entry 057's original validation protocol -- distinct from the deployed model itself, which is trained on all data for maximum production accuracy. The positive, meaningfully-sized correlation (0.553) between predicted uncertainty and actual error confirms the model knows when it's less confident, not just that it produces a number. Per-chip breakdown on the held-out set: Sherbrooke showed both the highest mean predicted std (1.05pts) and the highest MAE (1.56) -- the model correctly flags its own weakest chip.

Entry 072 — Re-Score and Deploy

All 24,003 precomputed qubit pairs across all three chips were re-scored with the deployed unified model, replacing both the old Entry 057 lookup (Kyiv/Sherbrooke, 2-chip-era) and the Entry 070 lookup (Brisbane, point-estimate-only) with one consistent table. The live demo's special-cased "no uncertainty band yet" messaging for Brisbane was removed -- there is no longer a code path where any chip lacks an uncertainty estimate. The validation-note footer was rewritten to honestly describe what this deployment model is (a generalist trained on data pooled from all three chips) versus what the separate leave-one-chip-out experiments (Entries 047-068) tested (true cold transfer to an entirely unseen chip) -- the two are not the same claim, and the demo now says so for every chip rather than singling out Brisbane.

Entries 071-072 — Conclusion

The project now has one deployed model, one validated uncertainty mechanism, and one honest validation note that applies uniformly across Kyiv, Sherbrooke, and Brisbane -- replacing two parallel model lineages with different guarantees. This closes the first item of the three-part roadmap agreed for this phase (uncertainty parity, a fourth never-seen-chip test, then productization); the fourth-chip generalization test is next.

Research Log — Index Addendum (Entries 071-072)

Files: entry071_mc_dropout.py, entry071_deploy_train.py, entry072_score_all_pairs.py. Data: quantumbridge_data/entry071_deploy_params.json (deployable weights, dropout_rate=0.15), entry072_lookup_with_uncertainty.json (rescored demo lookup, all 3 chips with calibrated uncertainty).